

CRUX

failure discovery → repair → qualification

for contact-rich robot manipulation — end to end on one AMD Radeon GPU

TASK SUCCESS 0/32 → 12/32 · +37.5 pp · exact McNemar p = 0.0005

replicates on an independent suite (0/32 → 9/32, p = 0.0039) · gate: APPROVED

virgin seeds 501-532 · zero discordant pairs against the candidate

THE DISCOVERY CAMPAIGN

18 matched sweep rounds

~1,100 batched episodes

9 mechanisms isolated

lift-then-translate transport

grip the connector link

re-observed grasp retries

the fingertip seat nudge

5 falsified families (1 was not)

THE METRIC BUG

Five seating methods all stalled

at 12.0-13.4 mm. That floor WAS

the seated position: the metric measured the connector's trailing joint, so success was impossible by design.

The nudge we called falsified five times had always worked.

SCALE, ON ONE RADEON

~200k-293k env-steps / second

4,096 batched environments

GPU 100% busy under live load

96-episode qualification: 114 s

Radeon PRO W7900 - gfx1100

ROCm 7.2.1 - Genesis 1.3.1

torch 2.13.0+rocm7.2

HONESTY IS THE PRODUCT

every number recomputed from

hash-verified raw episodes

a non-replicating claim was

withdrawn in writing

failed episodes never deleted

2 retractions, on the record

3 Genesis issues filed upstream

213 CPU tests - no GPU needed

VERIFY IT YOURSELF — CPU ONLY

```
git clone https://github.com/Enoch208/Crux && cd Crux && uv sync
```

```
uv run pytest -q # 213 tests, ~1 s
```

```
uv run crux validate evidence/manifest.json # 9/9 checks passed
```

Tamper with one byte of an episode file and it fails. That is the point.

[enoch208.github.io/Crux](https://github.com/Enoch208/Crux)

github.com/Enoch208/Crux · [docs/technical-report.md](#) · 3.5-min narrated demo in-repo